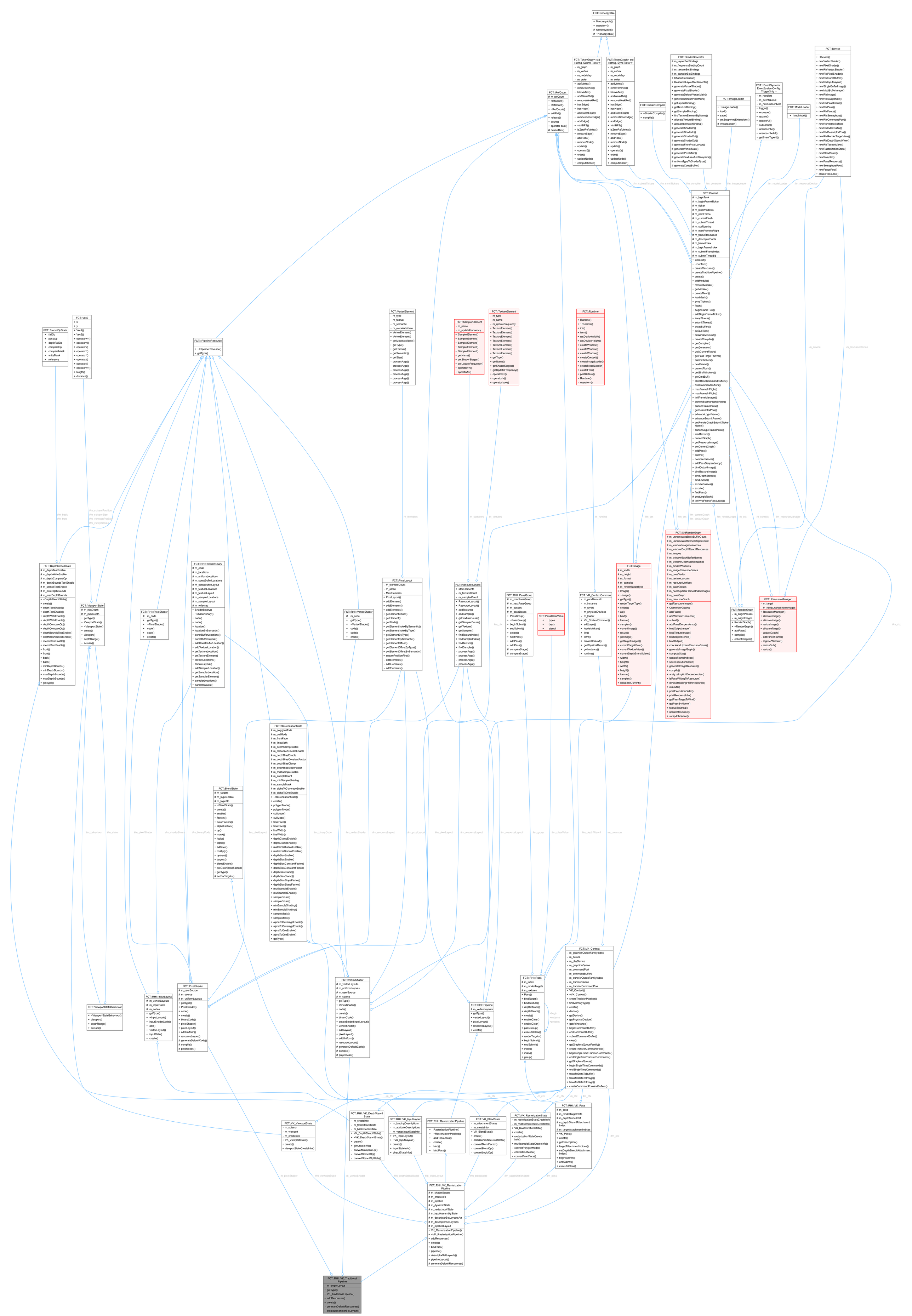
FCT::Noncopyable
+ Noncopyable()
+ operator=()
# Noncopyable()
# ~Noncopyable()
FCT::TokenGraph< std ::string, SubmitTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ addVertex()
+ removeVertex()
+ hasVertex()
+ addWeakRef()
+ removeWeakRef()
+ hasEdge()
+ hasNode()
+ addBoostEdge()
+ removeBoostEdge()
+ addEdge()
+ visitBFS()
+ isZeroRefVertex()
+ removeEdge()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
FCT::TokenGraph< std ::string, SyncTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ addVertex()
+ removeVertex()
+ hasVertex()
+ addWeakRef()
+ removeWeakRef()
+ hasEdge()
+ hasNode()
+ addBoostEdge()
+ removeBoostEdge()
+ addEdge()
+ visitBFS()
+ isZeroRefVertex()
+ removeEdge()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
FCT::RefCount
m_refCount
+ RefCount()
+ RefCount()
+ ~RefCount()
+ addRef()
+ release()
+ count()
+ operator bool()
# deleteThis()
FCT::ShaderCompiler
+ ~ShaderCompiler()
+ compile()
FCT::ShaderGenerator
m_layoutSetBindings
m_frequencyBindingCount
m_textureSetBindings
m_samplerSetBindings
+ ShaderGenerator()
+ ResourceLayoutToElements()
+ generateVertexShader()
+ generatePixelShader()
+ generateDefaultVertexMain()
+ generateDefaultPixelMain()
+ getLayoutBinding()
+ getTextureBinding()
+ getSamplerBinding()
+ findTextureElementByName()
+ allocateTextureBinding()
+ allocateSamplerBinding()
# generateShaderIn()
# generateShaderIn()
# generateShaderOut()
# generateShaderOut()
# generateFromPixelLayout()
# generateVertexMain()
# generatePixelMain()
# generateTexturesAndSamplers()
# uniformTypeToShaderType()
# generateConstBuffer()
FCT::ImageLoader
+ ~ImageLoader()
+ load()
+ save()
+ getSupportedExtensions()
# ImageLoader()
FCT::IEventSystem< EventSystemConfig:: TriggerOnly >
- m_handlers
- m_eventQueue
- m_nextSubscribeId
+ trigger()
+ enqueue()
+ update()
+ updateAll()
+ subscribe()
+ unsubscribe()
+ unsubscribeAll()
- getEventTypeId()
FCT::ModelLoader
+ loadModel()
FCT::Device
+ ~Device()
+ newVertexShader()
+ newPixelShader()
+ newRhiVertexShader()
+ newRhiPixelShader()
+ newRhiConstBuffer()
+ newRhiInputLayout()
+ newSingleBufferImage()
+ newMutilBufferImage()
+ newRhiImage()
+ newRhiSwapchain()
+ newRhiPassGroup()
+ newRhiPass()
+ newRhiFence()
+ newRhiSemaphore()
+ newRhiCommandPool()
+ newRhiVertexBuffer()
+ newRhiIndexBuffer()
+ newRhiDescriptorPool()
+ newRhiRenderTargetView()
+ newRhiDepthStencilView()
+ newRhiTextureView()
+ newRasterizationState()
+ newBlendState()
+ newSampler()
+ newPassResource()
+ newSemaphorePool()
+ newFencePool()
+ createResource()
#m_submitTickers
#m_syncTickers
#m_compiler
#m_generator
#m_imageLoader
#m_modelLoader
#m_resourceDevice
FCT::Context
m_logicTask
m_beginFrameTicker
m_ticker
m_bindWindows
m_nextFrame
m_currentFlush
m_submitThread
m_ctxRunning
m_maxFrameInFlight
m_frameResources
m_descriptorPools
m_frameIndex
m_logicFrameIndex
m_submitFrameIndex
m_submitThreadId
+ Context()
+ ~Context()
+ createResource()
+ createTraditionPipeline()
+ create()
+ addModule()
+ removeModule()
+ getModule()
+ createMesh()
+ loadMesh()
+ syncTickers()
+ flush()
+ beginFrameTick()
+ addBeginFrameTicker()
+ swapQueue()
+ submitThread()
+ swapBuffers()
+ defaultTick()
+ onWindowBound()
+ createCompiler()
+ getCompiler()
+ getGenerator()
+ waitCurrentFlush()
+ getPassTargetToWnd()
+ submitTickers()
+ nextFrame()
+ currentFlush()
+ getBindWindows()
+ getCmdBuf()
+ allocBaseCommandBuffers()
+ freeCommandBuffers()
+ maxFrameInFlight()
+ maxFrameInFlight()
+ initFrameManager()
+ currentSubmitFrameIndex()
+ currentFrameIndex()
+ getDescriptorPool()
+ advanceLogicFrame()
+ advanceSubmitFrame()
+ getRenderGraphSubmitTicker Name()
+ currentLogicFrameIndex()
+ loadTexture()
+ currentGraph()
+ getResourceImage()
+ setCurrentGraph()
+ addPass()
+ submit()
+ compilePasses()
+ addPassDependency()
+ bindOutputImage()
+ bindTextureImage()
+ bindDepthStencil()
+ bindOutput()
+ excutePasses()
+ excute()
+ findPass()
# postLogicTask()
# initWndFrameResources()
#m_ctx
#m_currentGraph
#m_defaultGraph
#m_renderGraph
#m_context
#m_resourceManager
-m_device
-m_resourceDevice
FCT::IPipelineResource
+ ~IPipelineResource()
+ getType()
FCT::VertexElement
- m_type
- m_format
- m_semantic
- m_modelAttribute
+ VertexElement()
+ VertexElement()
+ getModelAttribute()
+ getType()
+ getFormat()
+ getSemantic()
+ getSize()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
FCT::SamplerElement
- m_name
- m_updateFrequency
+ SamplerElement()
+ SamplerElement()
+ SamplerElement()
+ SamplerElement()
+ SamplerElement()
+ getName()
+ getShaderStages()
+ getUpdateFrequency()
+ operator==()
+ operator!=()
FCT::TextureElement
- m_type
- m_name
- m_updateFrequency
+ TexturElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ TextureElement()
+ getType()
+ getName()
+ getShaderStages()
+ getUpdateFrequency()
+ operator==()
+ operator!=()
+ operator bool()
FCT::Runtime
+ Runtime()
+ ~Runtime()
+ init()
+ term()
+ getDeviceWidth()
+ getDeviceHeight()
+ createWindow()
+ createWindow()
+ createWindow()
+ createContext()
+ createImageLoader()
+ createModelLoader()
+ createFont()
+ postUITask()
- Runtime()
- operator=()
FCT::StencilOpState
+ failOp
+ passOp
+ depthFailOp
+ compareOp
+ compareMask
+ writeMask
+ reference
FCT::Vec2
+ x
+ y
+ Vec2()
+ Vec2()
+ operator==()
+ operator+()
+ operator-()
+ operator*()
+ operator*()
+ operator/()
+ operator/()
+ operator==()
+ length()
+ distance()
FCT::DepthStencilState
m_depthTestEnable
m_depthWriteEnable
m_depthCompareOp
m_depthBoundsTestEnable
m_stencilTestEnable
m_minDepthBounds
m_maxDepthBounds
+ ~DepthStencilState()
+ create()
+ depthTestEnable()
+ depthTestEnable()
+ depthWriteEnable()
+ depthWriteEnable()
+ depthCompareOp()
+ depthCompareOp()
+ depthBoundsTestEnable()
+ depthBoundsTestEnable()
+ stencilTestEnable()
+ stencilTestEnable()
+ front()
+ front()
+ back()
+ back()
+ minDepthBounds()
+ minDepthBounds()
+ maxDepthBounds()
+ maxDepthBounds()
+ getType()
FCT::ViewportState
m_minDepth
m_maxDepth
+ getType()
+ ViewportState()
+ ~ViewportState()
+ create()
+ viewport()
+ depthRange()
+ scissor()
FCT::RHI::PixelShader
m_code
+ getType()
+ ~PixelShader()
+ code()
+ code()
+ create()
FCT::RHI::ShaderBinary
m_code
m_locations
m_uniformLocations
m_constBufferLocations
m_constBufferLayout
m_textureLocations
m_textureLayout
m_samplerLocations
m_samplerLayout
m_reflected
+ ShaderBinary()
+ ~ShaderBinary()
+ code()
+ code()
+ location()
+ locationBySemantic()
+ constBufferLocations()
+ constBufferLayout()
+ addConstBufferLocation()
+ addTextureLocation()
+ getTextureLocation()
+ getTextureElement()
+ textureLocations()
+ textureLayout()
+ addSamplerLocation()
+ getSamplerLocation()
+ getSamplerElement()
+ samplerLocations()
+ samplerLayout()
FCT::RHI::VertexShader
m_code
+ getType()
+ ~VertexShader()
+ code()
+ code()
+ create()
FCT::PixelLayout
- m_elementCount
- m_stride
- MaxElements
+ PixelLayout()
+ addElement()
+ addElements()
+ addElements()
+ getElementCount()
+ getElement()
+ getStride()
+ getElementIndexBySemantic()
+ getElementIndexByType()
+ getElementByType()
+ getElementBySemantic()
+ getElementOffset()
+ getElementOffsetByType()
+ getElementOffsetBySemantic()
+ ensurePositionFirst()
- addElements()
- addElements()
- addElements()
FCT::ResourceLayout
+ MaxElements
- m_textureCount
- m_samplerCount
+ ResourceLayout()
+ ResourceLayout()
+ addTexture()
+ addSampler()
+ getTextureCount()
+ getSamplerCount()
+ getTexture()
+ getSampler()
+ findTextureIndex()
+ findSamplerIndex()
+ findTexture()
+ findSampler()
- processArgs()
- processArgs()
- processArgs()
- processArgs()
FCT::RHI::PassGroup
m_prevPassGroup
m_nextPassGroup
m_passes
m_passDescs
+ PassGroup()
+ ~PassGroup()
+ beginSubmit()
+ endSubmit()
+ create()
+ nextPass()
+ addPass()
+ addPass()
# computeStage()
# computeStage()
FCT::PassClearValue
+ types
+ depth
+ stencil
FCT::VK_ContextCommon
- m_pickDeviceId
- m_instance
- m_layers
- m_physicalDevices
- m_loader
+ VK_ContextCommon()
+ addLayer()
+ loaderVulkan()
+ init()
+ term()
+ createContext()
+ getPhysicalDevice()
+ getInstance()
+ runtime()
FCT::Image
m_width
m_height
m_format
m_samples
m_renderTargetType
+ Image()
+ ~Image()
+ getType()
+ renderTargetType()
+ create()
+ as()
+ bind()
+ format()
+ samples()
+ currentImage()
+ resize()
+ getImage()
+ getTargetImages()
+ currentTargetView()
+ currentTextureView()
+ currentDepthStencilView()
+ width()
+ height()
+ width()
+ height()
+ format()
+ samples()
+ updateToCurrent()
FCT::OldRenderGraph
m_unnameWndBackBufferCount
m_unnameWndStencilDepthCount
m_windowImageResources
m_windowDepthStencilResources
m_images
m_windowBackBufferNames
m_windowDepthStencilNames
m_bindedWindows
m_imageResourceDescs
m_passVertex
m_textureLayouts
m_resourceVertices
m_passGroups
m_needUpdataFramesIndexImages
m_passGraph
m_resourceGraph
+ getResourceImage()
+ OldRenderGraph()
+ addPass()
+ addWindowResource()
+ submit()
+ addPassDependency()
+ bindOutputImage()
+ bindTextureImage()
+ bindDepthStencil()
+ bindOutput()
+ checkAndUpdateResourceSizes()
+ generateImageGraph()
+ computeSize()
+ updateFrameIndices()
+ saveExecutionOrder()
+ generateImageResource()
+ compile()
+ analyzeImplicitDependencies()
+ isPassWritingToResource()
+ isPassReadingFromResource()
+ execute()
+ printExecutionOrder()
+ printResourceInfo()
+ getPassTargetToWnd()
+ getPassByName()
+ formatToString()
+ updateResource()
+ swapJobQueue()
FCT::RenderGraph
- m_originPasses
- m_originImages
+ RenderGraph()
+ ~RenderGraph()
+ addPass()
+ compile()
- collectImages()
FCT::ResourceManager
- m_target
- m_needChangeIndexImages
+ ResourceManager()
+ allocateImage()
+ allocateImage()
+ resizeImage()
+ allocateTarget()
- updateGraph()
- addvanceFrame()
- registerWindow()
- resizeSub()
- resize()
#m_back
#m_front
#m_scissorPosition
#m_scissorSize
#m_viewportPosition
#m_viewportSize
-m_elements
-m_samplers
-m_textures
#m_ctx
-m_runtime
#m_ctx
#m_ctx
#m_currentGraph
#m_defaultGraph
#m_renderGraph
-m_ctx
-m_context
#m_resourceManager
#m_ctx
FCT::RasterizationState
m_polygonMode
m_cullMode
m_frontFace
m_lineWidth
m_depthClampEnable
m_rasterizerDiscardEnable
m_depthBiasEnable
m_depthBiasConstantFactor
m_depthBiasClamp
m_depthBiasSlopeFactor
m_multisampleEnable
m_sampleCount
m_minSampleShading
m_sampleMask
m_alphaToCoverageEnable
m_alphaToOneEnable
+ ~RasterizationState()
+ create()
+ polygonMode()
+ polygonMode()
+ cullMode()
+ cullMode()
+ frontFace()
+ frontFace()
+ lineWidth()
+ lineWidth()
+ depthClampEnable()
+ depthClampEnable()
+ rasterizerDiscardEnable()
+ rasterizerDiscardEnable()
+ depthBiasEnable()
+ depthBiasEnable()
+ depthBiasConstantFactor()
+ depthBiasConstantFactor()
+ depthBiasClamp()
+ depthBiasClamp()
+ depthBiasSlopeFactor()
+ depthBiasSlopeFactor()
+ multisampleEnable()
+ multisampleEnable()
+ sampleCount()
+ sampleCount()
+ minSampleShading()
+ minSampleShading()
+ sampleMask()
+ sampleMask()
+ alphaToCoverageEnable()
+ alphaToCoverageEnable()
+ alphaToOneEnable()
+ alphaToOneEnable()
+ getType()
FCT::BlendState
m_targets
m_logicEnable
m_logicOp
+ ~BlendState()
+ create()
+ enable()
+ factors()
+ colorFactors()
+ alphaFactors()
+ op()
+ mask()
+ logic()
+ alpha()
+ additive()
+ multiply()
+ opaque()
+ targets()
+ blendEnable()
+ srcColorBlendFactor()
+ getType()
# setForTargets()
#m_behaviour
#m_state
#m_pixelShader
#m_shaderBinary
#m_binaryCode
#m_pixelLayout
#m_binaryCode
#m_vertexShader
#m_resourceLayout
#m_pixelLayout
#m_pixelLayout
#m_resourceLayout
#m_resourceLayout
#m_group
#m_clearValue
#m_depthStencil
-m_common
#m_ctx
FCT::VK_ViewportState
- m_scissor
- m_viewport
- m_createInfo
+ VK_ViewportState()
+ create()
+ viewportStateCreateInfo()
FCT::RHI::VK_DepthStencil State
- m_createInfo
- m_frontStencilState
- m_backStencilState
+ VK_DepthStencilState()
+ ~VK_DepthStencilState()
+ create()
+ getCreateInfo()
- convertCompareOp()
- convertStencilOp()
- convertStencilOpState()
FCT::RHI::VK_InputLayout
- m_bindingDescriptions
- m_attributeDescriptions
- m_vertexInputStateInfo
+ VK_InputLayout()
+ ~VK_InputLayout()
+ create()
+ inputStateInfo()
+ pInputStateInfo()
FCT::RHI::RasterizationPipeline
+ RasterizationPipeline()
+ ~RasterizationPipeline()
+ addResources()
+ create()
+ bind()
+ bindPass()
FCT::VK_BlendState
- m_attachmentStates
- m_createInfo
+ VK_BlendState()
+ create()
+ colorBlendStateCreateInfo()
- convertBlendFactor()
- convertBlendOp()
- convertLogicOp()
FCT::VK_RasterizationState
- m_rasterizationStateCreateInfo
- m_multisampleStateCreateInfo
+ VK_RasterizationState()
+ create()
+ rasterizationStateCreate Info()
+ multisampleStateCreateInfo()
- convertPolygonMode()
- convertCullMode()
- convertFrontFace()
FCT::RHI::VK_Pass
m_desc
m_renderTargetRefs
m_depthStencilRef
m_depthStencilAttachment Index
m_targetAttachmentIndices
+ VK_Pass()
+ create()
+ getDescription()
+ targetAttachmentIndices()
+ setDepthStencilAttachment Index()
+ beginSubmit()
+ endSubmit()
+ executeClear()
-m_pixelShader
#m_viewportState
-m_vertexShader
#m_depthStencilState
#m_inputLayout
#m_blendState
#m_rasterizationState
#m_pass
#m_ctx
FCT::RHI::VK_Rasterization Pipeline
m_shaderStages
m_createInfo
m_pipeline
m_dynamicState
m_vertexInputState
m_inputAssemblyState
m_descriptorSetLayoutsArr
m_descriptorSetLayouts
m_pipelineLayout
+ VK_RasterizationPipeline()
+ ~VK_RasterizationPipeline()
+ addResources()
+ create()
+ bindPass()
+ pipeline()
+ descriptorSetLayouts()
+ pipelineLayout()
# generateDefaultResources()
FCT::VertexShader
m_vertexLayouts
m_uniformLayouts
m_userSource
m_source
+ getType()
+ VertexShader()
+ code()
+ create()
+ binaryCode()
+ createBindedInputLayout()
+ vertexShader()
+ addLayout()
+ pixelLayout()
+ addUniform()
+ resourceLayout()
# generateDefaultCode()
# compile()
# preprocess()
FCT::RHI::Pipeline
m_vertexLayouts
+ getType()
+ vertexLayout()
+ pixelLayout()
+ resourceLayout()
+ create()
FCT::RHI::Pass
m_index
m_renderTargets
m_textures
+ Pass()
+ bindTarget()
+ bindTexture()
+ depthStencil()
+ depthStencil()
+ create()
+ enableClear()
+ enableClear()
+ passGroup()
+ executeClear()
+ renderTargets()
+ beginSubmit()
+ endSubmit()
+ index()
+ index()
+ group()
+begin
+external
+present
FCT::VK_Context
- m_graphicsQueueFamilyIndex
- m_device
- m_phyDevice
- m_graphicsQueue
- m_commandPool
- m_commandBuffers
- m_transferQueueFamilyIndex
- m_transferQueue
- m_transferCommandPool
+ VK_Context()
+ ~VK_Context()
+ createTraditionPipeline()
+ findMemoryType()
+ create()
+ device()
+ getDevice()
+ getPhysicalDevice()
+ getVkInstance()
+ beginCommandBuffer()
+ endCommandBuffer()
+ submitCommandBuffer()
+ clear()
+ getGraphicsQueueFamily()
+ createTransferCommandPool()
+ beginSingleTimeTransferCommands()
+ endSingleTimeTransferCommands()
+ getGraphicsQueue()
+ beginSingleTimeCommands()
+ endSingleTimeCommands()
+ transferDataToBuffer()
+ transferDataToImage()
+ transferDataToImage()
- createCommandPoolAndBuffers()
FCT::RHI::InputLayout
m_vertexLayouts
m_inputRates
m_codes
+ getType()
+ ~InputLayout()
+ inputShaderCode()
+ add()
+ vertexLayout()
+ inputRate()
+ create()
FCT::PixelShader
m_userSource
m_source
m_uniformLayouts
+ getType()
+ PixelShader()
+ code()
+ create()
+ binaryCode()
+ pixelShader()
+ pixelLayout()
+ addUniform()
+ resourceLayout()
# generateDefaultCode()
# compile()
# preprocess()
FCT::ViewportStateBehaviour
+ ~ViewportStateBehaviour()
+ viewport()
+ depthRange()
+ scissor()
FCT::RHI::VK_Traditional Pipeline
- m_emptyLayout
+ getType()
+ VK_TraditionalPipeline()
+ addResources()
+ create()
- generateDefaultResources()
- createDescriptorSetLayouts()